Q1. Refer to the circuit show in Figure Q1

(i) Use nodal analysis to calculate the v_x indicated in the circuit

(ii) Employ superposition to determine the value of v_x arising from the 5V voltage source only.

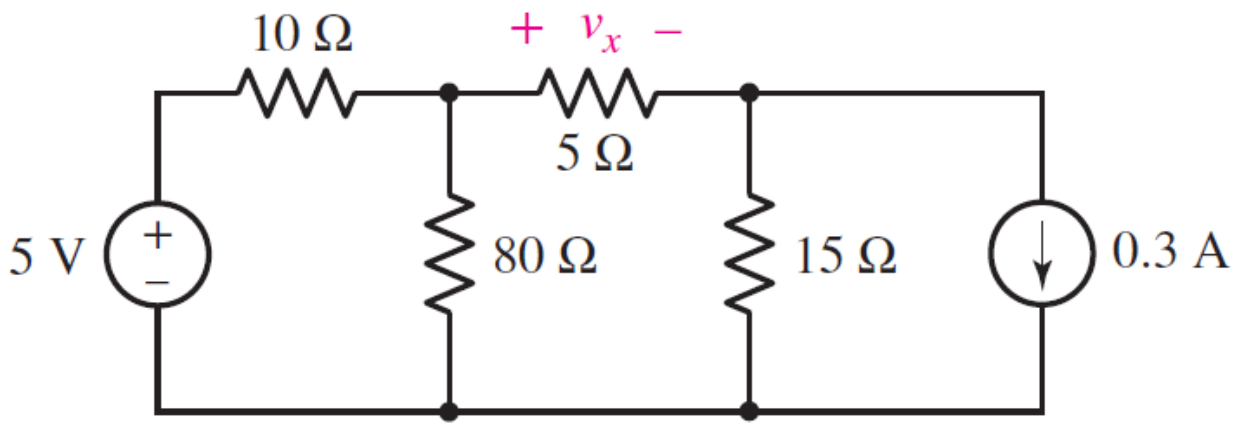
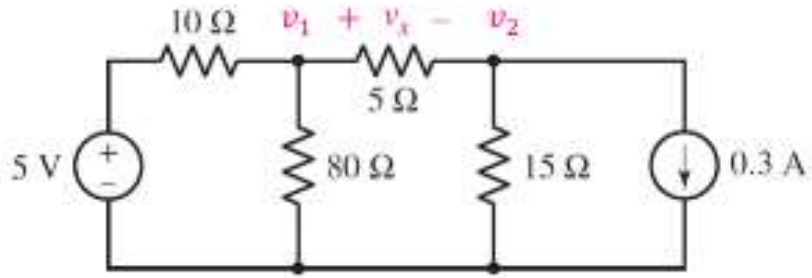


Figure Q1

(i) Chose the bottom node as reference node and assign nodal voltage v1 v2 to the left and right node.

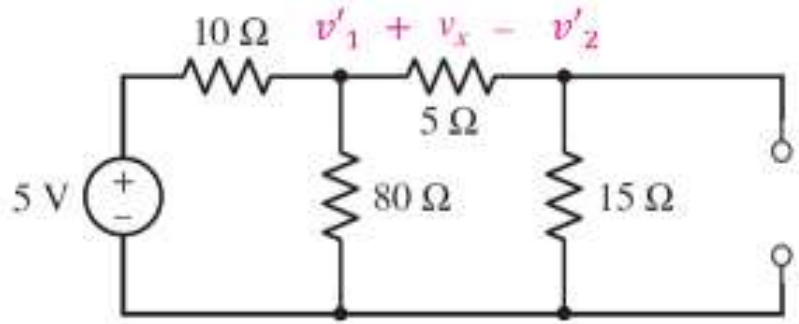


$$\frac{v_1 - 5}{10} + \frac{v_1}{80} + \frac{v_1 - v_2}{5} = 0$$

$$\frac{v_2 - v_1}{5} + \frac{v_2}{15} = -0.3$$

Solving, v_1=1.6923 V and v_2=0.1442 V
The unknown $v_x = v_1 - v_2 = 1.5481$ V

(ii) turn off the current source by replacing it to an open circuit



Apply KCL at node one

$$\frac{v'_1-5}{10}+\frac{v'_1}{80}+\frac{v'_1}{20}=0$$

v'_1=3.076V, v'_2= 15/(15+5)* v'_1=2.307V

$v'_x = v'_1 - v'_2 = 0.768V$

Q2. Consider the circuit in Figure Q2. i_1, i_2, i_3 are the mesh currents in mesh 1, mesh 2 and mesh 3, respectively. Use mesh analysis/supermesh techniques to calculate the mesh current i_1, i_2, and i_3.

Calculate the power delivered or absorbed by each independent voltage source and identify which source is delivering power and which sources is absorbing power.

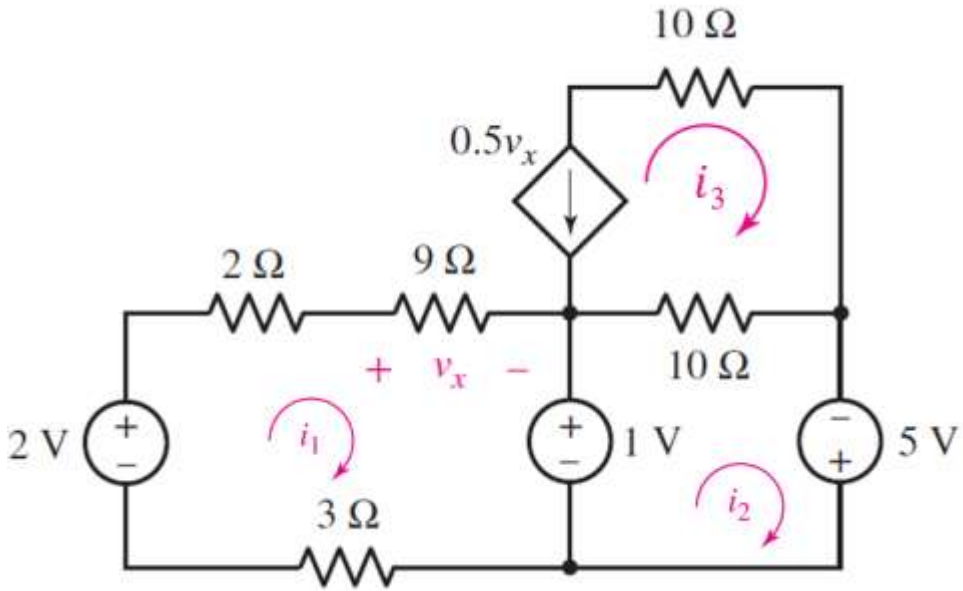


Figure Q2

(i) In mesh 1 we have

$-2 + (2 + 9 + 3)i_1 + 1 = 0$

which can be solved for $i_1 = 71.43$ mA

In mesh 2 we have

$-1 + 10i_2 - 10i_3 - 5 = 0$

➔ $i_2 = 278.6$ mA

By inspection, $i_3 = -0.5v_x = -0.5(9i_1) = -321.4$ mA

(ii) The power absorbed by the 2V source is

$P_{2V} = -i_1(2) = -142.86$ mW

$P_{1V} = (i_1 - i_2) * 1 = -207.17$ mW

$P_{5V} = -i_2(5) = -1.39$

They are all delivering power

Q3. Consider the circuit in Figure Q3.

(a) Determine the Norton equivalent circuit with respect to terminal *a* and terminal *b* (i.e., drawing the Norton equivalent circuit and calculating R_N and I_N).

(b) Suppose that load resistor R_L is connected between terminal *a* and terminal *b*. What's the value of R_L that will result in the maximum power transfer to the load resistor? What is the maximum power dissipation in R_L.

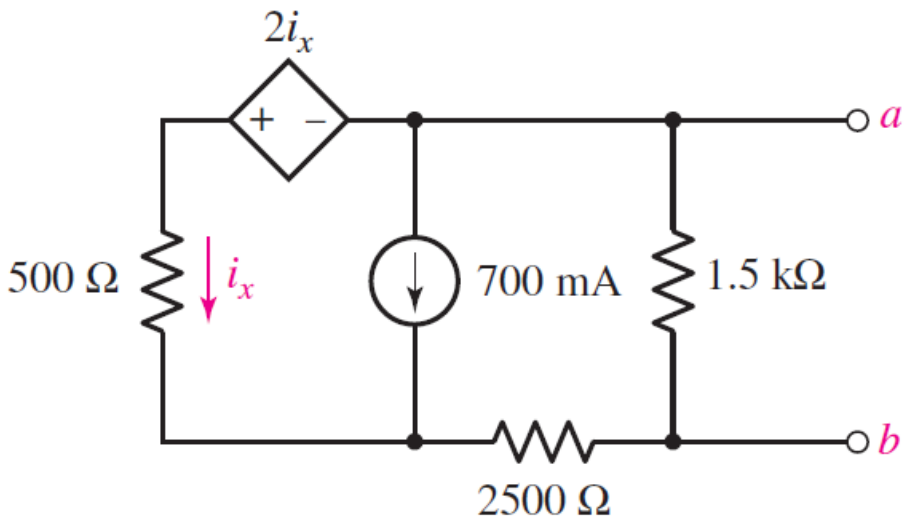
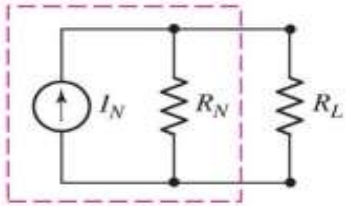
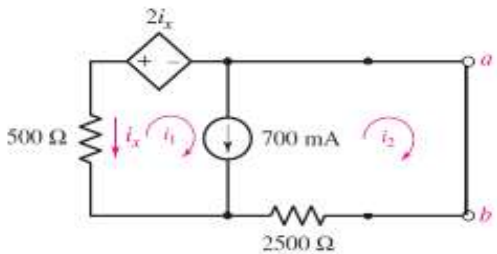


Figure Q3

(i) Draw the Norton equivalent circuit



We short the terminals of the network and compute the short circuit current. To do this, define two clockwise mesh currents (the 1500 Ω resistor is shorted out).



Apply supermesh by combining mesh 1 and mesh 2

$500i_1 + 2i_x + 2500i_2 = 0$

$i_1 - i_2 = 0.7$

$i_1 = -i_x$

Solving, $i_2 = i_{sc} = i_N = -116.3$ mA

Since we cant calculate the R_N directly we can calculate Voc instead use mesh analysis (other method also ok):

$500i_1 + 2i_x + 1500i_2 + 2500i_2 = 0$

$i_1 - i_2 = 0.7$

$i_1 = -i_x$

i_2=-0.0775A

Voc=i2*1500=-116.3V

$R_{TH} = \frac{Voc}{I_N} = 1000\ \Omega$

(ii) When $R_L = R_{TH} = 1000\ \Omega$ get the maximum power transfer to R_L

$P_{max} = \frac{V_{TH}^2}{4R_{TH}} = 3.38\text{W}$ (or $P_{max} = \frac{I_N^2 R_N}{4} = 3.38\text{W}$